

DHCP Server

Linux System & Network Administration

1. Service Fundamentals

DHCP (Dynamic Host Configuration Protocol) automatically assigns IP addresses and network parameters (subnet mask, gateway, DNS, lease time) to clients. It eliminates manual IP configuration and prevents address conflicts.

Item	Details
Service / Package	DHCP · isc-dhcp-server
Transport / Ports	UDP · Server: 67 · Client: 68
Main Config	/etc/dhcp/dhcpd.conf
Interface Binding	/etc/default/isc-dhcp-server
Lease Database	/var/lib/dhcp/dhcpd.leases
Logs	/var/log/syslog (grep dhcpd)
Service Unit	isc-dhcp-server.service

2. Architecture & DORA Process

The DHCP server maintains an IP pool and responds to client requests. Clients obtain configuration through the four-step DORA exchange.

Step	Message	Direction	Purpose
1	DHCPDISCOVER	Client → Broadcast	Client seeks any DHCP server
2	DHCPOFFER	Server → Client	Server offers IP + options
3	DHCPREQUEST	Client → Server	Client requests the offered IP
4	DHCPACK	Server → Client	Server confirms the assignment

Lease lifecycle: Assigned → Active → Renewed (or) Expired → Returned to pool. Client may also send DHCPRELEASE voluntarily.

3. Lab Network Plan

Parameter	Value
Network	192.168.100.0/24
Server Static IP	192.168.100.10 (outside the pool)
Client Pool	192.168.100.101 – 192.168.100.150
Interface	enp0s9 (Internal Network)
Adapters	Host-Only (admin) · NAT (internet) · Internal (DHCP)

4. Installation

Update package list and install the DHCP server:

Command:

```
sudo apt update
```

Command:

```
sudo apt install isc-dhcp-server
```

Identify the network interface that will serve DHCP:

Command:

```
ip a
```

In this lab the interface is **enp0s9**. Confirm it is up and has the static address 192.168.100.10.

5. Configuration

5.1 Bind DHCP to the correct interface

Open the interface configuration file:

Command:

```
sudo nano /etc/default/isc-dhcp-server
```

Find the line INTERFACESv4 and set it to:

```
INTERFACESv4="enp0s9"
```

Save and exit the editor.

5.2 Configure the subnet and IP pool

Open the main DHCP configuration file:

Command:

```
sudo nano /etc/dhcp/dhcpd.conf
```

Add the following configuration:

```
authoritative;
subnet 192.168.100.0 netmask 255.255.255.0 {
    range 192.168.100.101 192.168.100.150;
}
```

***authoritative**; marks this server as the official DHCP server for the network. Keep the server static IP (192.168.100.10) outside the dynamic range.*

5.3 Recommended global options

Add these lines to the same dhcpd.conf file:

```
default-lease-time 86400;
max-lease-time 172800;
ddns-update-style none;
option domain-name-servers 8.8.8.8, 1.1.1.1;
option routers 192.168.100.1;
option broadcast-address 192.168.100.255;
option domain-name "lab.local";
```

5.4 Optional — MAC-based reservation

To give a specific client a fixed IP based on its MAC address:

```
host client1 {
    hardware ethernet aa:bb:cc:dd:ee:ff;
    fixed-address 192.168.100.50;
}
```

5.5 Validate, restart and enable the service

Check configuration syntax for errors:

Command:

```
sudo dhcpd -t -cf /etc/dhcp/dhcpd.conf
```

Restart the service to apply changes:

Command:

```
sudo systemctl restart isc-dhcp-server
```

Enable the service to start on boot:

Command:

```
sudo systemctl enable isc-dhcp-server
```

6. Verification

Check service status (must show active/running):

Command:

```
sudo systemctl status isc-dhcp-server
```

Confirm the server is listening on UDP port 67:

Command:

```
sudo ss -ltnp | grep :67
```

Monitor live DHCP activity while a client requests an address:

Command:

```
sudo tail -f /var/log/syslog | grep dhcpd
```

View current leases:

Command:

```
sudo cat /var/lib/dhcp/dhcpd.leases
```

From a client on the same Internal Network, request an address. It should receive an IP in the range 192.168.100.101–150 along with the configured gateway and DNS options.

7. Configuration Reference

File / Directive	Purpose
/etc/dhcp/dhcpd.conf	Main configuration file
/etc/default/isc-dhcp-server	Interface binding (INTERFACESv4)
/var/lib/dhcp/dhcpd.leases	Lease database
authoritative;	Declare this server authoritative
subnet ... { }	Define the managed network
range start end;	Dynamic IP pool
option routers ...;	Default gateway for clients
option domain-name-servers ...;	DNS servers for clients
host { hardware ethernet ...; }	MAC-based fixed reservation

8. Troubleshooting

Problem	Check	Solution
Service fails to start	dhcpd -t · logs	Fix syntax errors; correct INTERFACESv4
Client gets no IP	Same network? · ss · logs	Both on Internal Network; open UDP 67/68
Wrong interface	/etc/default/isc-dhcp-server	Set INTERFACESv4 to the correct NIC
Pool exhausted	dhcpd.leases	Expand range or reduce lease time
Config ignored	Service not restarted	Run dhcpd -t then systemctl restart

9. Security

- Bind DHCP only to trusted interfaces (INTERFACESv4).
- Use short lease times on guest networks.
- Use MAC reservations for critical devices.
- Allow UDP 67/68 only from the intended subnet on the firewall.
- Never run DHCP on internet-facing interfaces.

10. Real-World Scenario

A company has three departments on separate subnets. A central DHCP server (or DHCP relay) serves distinct pools: HR 192.168.10.101–150, Finance 192.168.20.101–150, IT 192.168.30.101–150. Clients receive addresses only from their own department pool.

11. Essential Commands

Command	Purpose
sudo apt install isc-dhcp-server	Install DHCP server
sudo dhcpd -t -cf /etc/dhcp/dhcpd.conf	Check configuration syntax
sudo systemctl restart isc-dhcp-server	Apply configuration changes
sudo systemctl status isc-dhcp-server	Check if service is running
sudo ss -ltnp grep :67	Verify listening port
sudo tail -f /var/log/syslog grep dhcpd	Live log monitoring
sudo cat /var/lib/dhcp/dhcpd.leases	View current leases

12. Important Points (Interview / Viva)

1. DHCP automatically assigns IP addresses and network parameters to clients.
2. Ports: UDP 67 (server), UDP 68 (client).
3. DORA = Discover → Offer → Request → Acknowledgment.
4. Main config file: /etc/dhcp/dhcpd.conf · Package: isc-dhcp-server.
5. Validate config with: sudo dhcpd -t -cf /etc/dhcp/dhcpd.conf
6. Leases are stored in /var/lib/dhcp/dhcpd.leases
7. Reservation = fixed IP for a specific MAC address.